

Assay Lab Requirements

What we saw on site, what we understood you to need,
and what we still have to ask before anyone can be held to
an estimate.

Prepared for	laboratory management, geology and IT
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Status	draft, awaiting confirmation
Contents	70 requirements, 12 assumptions, 20 open questions

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Why this document exists, and what we need you to do with it

We spent a day in the laboratory being walked through the process, from samples arriving out of the field through weighing, preparation and splitting, fire assay, and instrumental finish, to the results that leave the laboratory. This document records what we understood, in the form of numbered requirements, so that the understanding can be corrected before it becomes software.

Nothing here is a criticism of how the laboratory works today. The process we observed is sound; what it lacks is a system underneath it that carries the numbers without a person retyping them.

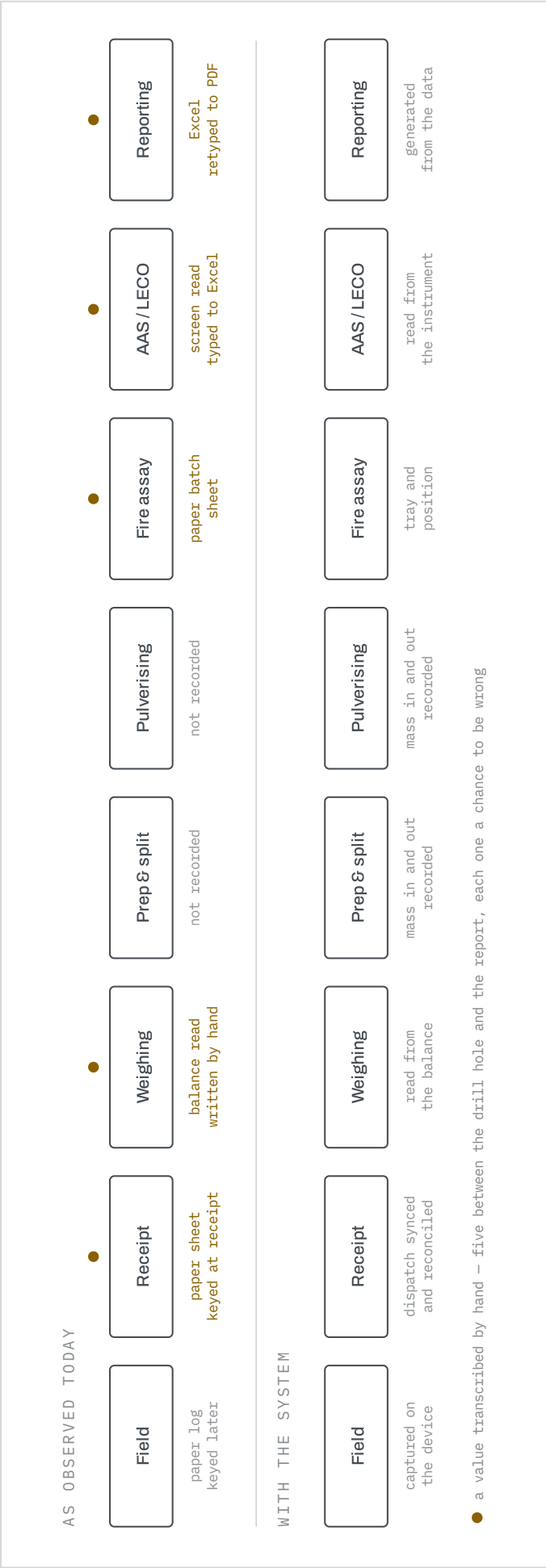
What we are asking of you: read the requirements in section 5 and mark any that are wrong, missing, or worded in a way you would not sign your name to. Pay particular attention to the ones marked *inferred* — those are our interpretation rather than something you told us, and they are the ones most likely to be wrong. Then look at the assumptions in section 7 and the questions in section 8, because those are what currently stand between this document and a firm estimate.

Requirement numbers are permanent. If a requirement is removed it will be marked as withdrawn rather than renumbered, so that a reference made today still means the same thing in six months.

The process as we observed it

Samples arrive from the field with a paper record of where they came from. They are received, weighed, and prepared — dried, crushed, split and pulverised — and a portion is weighed out into a batch of fifty for fire assay. After fusion and cupellation the prill is weighed, and the finish is read on the atomic absorption spectrometer. Sulphur determination runs separately on the LECO analyser. Results are assembled in Excel and issued as a branded PDF. Where doré or bullion is poured, bars are sampled and assayed through to a purity figure.

Each instrument is attached to a PC. At present the number those instruments produce is read by a person and written or typed somewhere else. That is the single observation from which most of this document follows.



Where the numbers are handled today, and where they would come from instead. The stages do not change – the laboratory process is not being redesigned. What changes is that a value is captured once, at the instrument that produced it, and is never retyped afterwards.

What we found

Six observations that shape everything in section 5.

- **Numbers are transcribed by hand at five points** between the drill hole and the issued report. Each one is an opportunity for a transposed digit, and in gold assay a transposed digit can send ore to the waste dump.
- **There is no single place to ask where a sample is.** The answer currently comes from whoever last handled it. This was raised first and most often on the visit.
- **Field data and laboratory data are not connected.** Hole and depth information is recorded on paper by geology and re-entered later, so the laboratory cannot check what it received against what was sent.
- **The instruments are already networked,** which is the single most helpful fact we found. Direct capture is a matter of software rather than of new cabling or new equipment.
- **Quality control is performed but not enforced.** Controls are run; whether a batch is held when one fails depends on a person noticing.
- **Records live in spreadsheets and on paper.** They are legible and they are current, but they cannot show who changed a number, when, or what it was before — which is precisely what an assessor, an investor or a disputed result requires.

What the system is, in one paragraph

A laboratory information management system covering the chain from a sample collected in the field to a purity figure on a poured bar. It captures readings directly from the balances, the atomic absorption spectrometer and the LECO analyser; tracks every sample and every transformation it undergoes; batches work into trays with defined quality control positions; enforces configurable control rules; records who did what and when in a record that cannot be quietly altered; and issues the Excel and PDF reports the laboratory issues today. It runs on site and keeps running when the external connection does not.

Requirements

Seventy requirements in twelve groups. Priority describes the first release: *Must* means the system is not usable without it, *Should* means it belongs in the first release if the schedule allows, and *Later* means it is agreed but deferred.

observed – we saw this in the laboratory **stated** – you told us this
inferred – our interpretation, please confirm

A Sample lifecycle and tracking

R-01	Every sample carries a unique identifier generated by the system. The format is configurable and includes a check digit, so a mistyped or misread number fails immediately rather than attaching a result to the wrong sample.	MUST inferred
R-02	Sample receipt records date and time, the receiving officer, the dispatch it belongs to, and the condition of the samples as received.	MUST observed
R-03	Every physical transformation of a sample is recorded as a linked parent and child, carrying mass in, mass out, operator and time. Covers drying, crushing, splitting, pulverising and weighing out the assay charge.	MUST observed
R-04	A user with the appropriate role can determine the stage and physical location of any sample at any moment. Stated on the visit as the ability to follow a sample through the entire value chain.	MUST stated
R-05	Sample status updates automatically as work is recorded. There is no separate manual step to change status. A status that has to be set by hand will be wrong within a week.	MUST inferred
R-06	Retained pulps and rejects are tracked as retained material with a storage location.	SHOULD inferred
R-07	The system supports composite samples formed from more than one parent sample.	SHOULD inferred
R-08	Samples can be received and logged before their field data has arrived. Receipt creates provisional records that reconcile when the dispatch arrives. Missing field data must never stop the laboratory.	MUST inferred
R-09	Receipt produces a reconciliation report: expected against received, missing, unexpected and mismatched identifiers. This is the check currently performed by eye against a paper sheet.	MUST inferred

B Field data capture

R-10	Geologists record hole identifier, from and to depth, sample number, sample type, date and collector at the point of collection.	MUST stated
R-11	Field capture functions with no network connectivity of any kind.	MUST stated

R-12	Field data transfers to the laboratory system over the local network when the device reaches site. No internet connection is involved; the device and the laboratory server are on the same local network.	MUST stated
R-13	Where a device cannot transfer, a printed dispatch manifest carrying a machine-readable code can be scanned at receipt. Fallback channel, so that a failure of the primary channel is an inconvenience rather than a stoppage.	SHOULD inferred
R-14	Geology can insert its own control samples into a dispatch and set whether the laboratory is able to identify them.	MUST stated
R-15	The identity of a blind control is not disclosed to laboratory users through any screen, report, export, alert or message until the configured reveal event. Enforced in the data itself rather than by hiding fields on screen.	MUST inferred
R-16	Dispatch data can be entered at the laboratory where field devices are not yet in use. Geology records on paper today. The laboratory must gain value before field adoption completes.	MUST observed

c Instrument integration

R-17	Weights are captured directly from the balances. No operator types a weight. Identified on the visit as the single most important requirement.	MUST stated
R-18	Results are captured directly from the atomic absorption spectrometer.	MUST stated
R-19	Results are captured directly from the LECO combustion analyser. Used for sulphur and, where applicable, carbon determination.	MUST stated
R-20	Capture supports serial connection, exported files and print streams, so that instruments of different ages can be connected.	MUST inferred
R-21	Every captured reading is bound to a specific sample or tray position, and the operator confirms the pairing before it is committed. Prevents the common failure of fifty correct readings against forty-nine correct samples.	MUST inferred
R-22	Raw instrument output is retained exactly as received and can be re-processed later.	MUST inferred
R-23	Manual entry with double verification remains available for any instrument that cannot be connected.	MUST inferred

R-24	A new instrument model can be added by configuration rather than by changing the software, wherever its output format allows.	SHOULD inferred
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R-25	Instrument identity, method and operator are recorded against every reading.	MUST inferred
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D Fire assay workflow and batching

R-26	Samples are grouped into batches. Batch size is configurable; the current standard is fifty.	MUST observed
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R-27	A batch is a tray of numbered positions, and the occupant of every position is recorded. Position is retained because furnace behaviour is not uniform across a tray and position-related bias is a real diagnostic signal.	MUST inferred
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R-28	The positions, types and frequency of quality control inserts are defined by a configurable batch template.	MUST stated
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R-29	Fusion, cupellation, prill weighing and parting are recorded as distinct steps with operator and time.	MUST observed
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R-30	Furnace identity and, where the laboratory tracks them, flux and cupel lots are recorded against a batch.	SHOULD inferred
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R-31	The assay charge mass is captured from the balance rather than entered.	MUST stated
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E Quality control

R-32	Quality control rules are configurable by a user holding the appropriate rights. Covers reference material tolerance, blank limit, duplicate difference limit and insertion rate.	MUST stated
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R-33	Certified values and standard deviations for reference materials are held per lot, taken from the lot certificate.	MUST inferred
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R-34	Control results are evaluated automatically as they are captured, and failures are raised without anyone having to look for them.	MUST inferred
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R-35	Control charts are available per reference material lot and per element over time.	SHOULD inferred
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R-36	The action taken on a control failure is configurable: warn, hold the batch, or quarantine it.	MUST stated
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R-37	Batch release requires approval by a user holding the approver role, and the person who produced a result cannot approve it. Approval currently rests with the laboratory manager.	MUST stated
R-38	Re-assays are recorded against the original result, flagged as re-assays, and shown as such in reports. Stated explicitly on the visit.	MUST stated
R-39	Laboratory controls and submitter controls are evaluated separately. Submitter controls do not gate laboratory release. A blind control cannot hold a batch, because the hold would reveal it.	MUST inferred

F Traceability and audit

R-40	Every action is logged with the user, their role, the date and time, the action taken, and the values before and after.	MUST stated
R-41	Audit records cannot be altered or removed by any user, including administrators, and any tampering is detectable.	MUST inferred
R-42	Results are never overwritten. A correction creates a new version carrying a reason, with the previous value still visible.	MUST inferred
R-43	Approval is captured as an electronic signature bound to the exact version of the data approved.	MUST inferred
R-44	The full history of any sample, batch or bar can be produced as a report for an auditor.	MUST inferred
R-45	Changes to configuration are themselves audited and versioned, and past evaluations remain reproducible. Without this, widening a tolerance today would change whether a batch passed last year.	MUST inferred
R-46	Records are retained for a defined period, set by configuration. Period to be confirmed. See Q-17.	MUST stated

G Doré and bullion

R-47	Melts and bars are recorded with serial number, gross mass and the melt they came from.	MUST stated
R-48	A bar sample re-enters the assay chain and produces a fineness recorded against that bar.	MUST stated

R-49	A certificate can be produced for a bar, showing purity and the assay evidence supporting it.	MUST inferred
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R-50	Where bars are sent to an external refiner, the refiner's return assay can be recorded and compared against the laboratory's own figure. The variance between the two is money, and is currently reconciled outside any system.	LATER inferred
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H Reporting and outputs

R-51	Reports are produced as Excel workbooks and as branded PDFs on company letterhead. Matches what the laboratory issues today.	MUST observed
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R-52	Report templates and branding are configurable without changing the software.	SHOULD inferred
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R-53	A re-issued report supersedes its predecessor visibly and carries the reason for re-issue.	MUST inferred
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R-54	Results can be exported in a format the geological database can import without re-typing.	MUST inferred
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R-55	Dashboards show work in progress, turnaround time, and batches awaiting approval.	SHOULD inferred
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R-56	A shift handover summary can be produced automatically.	LATER inferred
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I Users, roles and access

R-57	Access is controlled by role, covering field collection, receipt, preparation, assay, quality control, approval, read-only and administration.	MUST stated
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R-58	Every user authenticates individually. Shared logins are not supported, because they would defeat the audit trail.	MUST inferred
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R-59	The system supports the current user population of approximately ten to fifteen, with no barrier to growth.	MUST stated
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R-60	Screens used at the bench are usable while wearing gloves and protective equipment. Large targets, high contrast, no controls that depend on hovering.	SHOULD observed
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J Configuration and reuse

R-61	Batch templates, quality control rules, identifier schemes, workflow stages, report templates and branding are all configuration rather than code.	MUST stated
R-62	The system can be deployed for a different laboratory, including a third-party laboratory, without modifying the software. Stated as a requirement so that the product is resellable.	MUST stated
R-63	Configuration is versioned, and results remain bound to the configuration version under which they were produced.	MUST inferred

K Data migration

R-64	Historical results held in Excel and CSV can be imported, with column mapping, validation, a dry run and a reconciliation report.	MUST stated
R-65	Imported records are identified as migrated, retain a reference to their source file, and are read-only. They have no instrument capture, audit trail or control evaluation behind them and must not be mistaken for records the system produced.	MUST inferred

L Environment and availability

R-66	The system operates in full when the site has no external connectivity.	MUST stated
R-67	Where a central tier is used, site data transfers to it when connectivity is available, with no loss of records.	MUST stated
R-68	Laboratory operation continues if the wider corporate network is unavailable.	MUST inferred
R-69	Backups run automatically and restoration is tested, not assumed.	MUST inferred
R-70	The deployment model, on-premise or hosted, is to be confirmed by the client's IT function. The design does not depend on the answer. See Q-16.	MUST inferred

06

Not in the first release

Recorded so that their absence is a decision rather than an oversight.

- Tracking individual crucibles and cupels through their firing life. Deferred by agreement; reference material lots remain in the first release.
- Sending worklists to instruments. The first release reads results from instruments but does not drive them.
- Plant-wide metal accounting and mass balance reconciliation. This crosses into processing systems and is a separate conversation.
- Integration with corporate ERP systems.
- Client, job and invoicing handling for third-party laboratory work.
- Replacing acQuire or any geological database. The system exchanges data with it; it does not supersede it.
- Predictive or anomaly detection analytics, which require a period of the system's own clean data before they can mean anything.

07

Assumptions

Where we did not have an answer we have assumed one and written down what happens if the assumption turns out to be wrong. Correcting any of these now is far cheaper than discovering it during the build.

ID	ASSUMPTION	IF IT IS WRONG
A-01	This is an in-house mine laboratory. Work for external clients is not in the first release, though the design allows for it.	Third-party client handling, quotations and invoicing would become in-scope work.
A-02	Fifty is the standard fire assay tray size; other sizes are handled by configuration rather than being unsupported.	None material. Configuration absorbs it.
A-03	Instrument PCs run a version of Windows capable of hosting a small capture service.	Older or locked-down machines fall back to file-based capture, which is slower to build and covers fewer instruments.
A-04	IT will permit installation of the capture service and provide the necessary administrator access.	Direct capture becomes impossible for the affected instruments and manual entry with verification is the fallback. This would remove the largest single benefit of the project.
A-05	Assay results must reach acQuire. The laboratory system is the authority on laboratory work; acQuire remains the authority on hole and depth data.	If the laboratory is expected to become the master for geological data, the scope changes substantially.
A-06	The laboratory is not presently accredited to ISO/IEC 17025. The system is designed to support accreditation regardless.	If accreditation is already held or imminent, validation and documentation effort increases.

ID	ASSUMPTION	IF IT IS WRONG
A-07	There is one laboratory site at present, and the design allows for more.	None immediate. Multi-site brings the central tier forward in the plan.
A-08	Records are retained for a period measured in years, with the exact period to be confirmed.	Storage sizing and archive design change.
A-09	Approval authority rests with the laboratory manager.	Role definitions change; the principle of separating production from approval does not.
A-10	No barcode printers or scanners are in use today. Their introduction is recommended but not assumed by the design.	None. The system works with typed identifiers and improves when scanners arrive.
A-11	Site power is stable enough to run a server protected by an uninterruptible power supply.	Hardware specification and the backup regime change.
A-12	Doré or bullion is produced on site and assayed to a purity figure.	If bars leave unassayed for a third-party refiner, refiner reconciliation moves up in priority and on-site bar certification falls away.

08

What we still need

Twenty items, with the person best placed to answer each. The five marked **BLOCKING** prevent a firm estimate being given — not because the design depends on them, but because the effort does.

Items Q-01 to Q-03 are covered by the instrument survey issued alongside this document. It is a form rather than a conversation, because the detail it asks for is easier to read off the equipment than to recall in a meeting.

ID	WHAT WE NEED	OWNER	PRIORITY	WHY IT MATTERS
Q-01	The make, model, vendor software and software version of every balance, the AAS and the LECO analyser.	Laboratory / IT	BLOCKING	Integration effort cannot be estimated without it. It is the largest unknown in the project.
Q-02	How each instrument makes its data available today: serial output, an exported file, a printout, or a reading taken off the screen.	Laboratory	BLOCKING	Determines whether an instrument is two weeks of work or two months.
Q-03	The Windows version on each instrument PC, and who holds administrator rights.	IT	BLOCKING	Decides whether the capture service can be installed at all.

ID	WHAT WE NEED	OWNER	PRIORITY	WHY IT MATTERS
Q-04	Sample throughput: per shift, per day, and in the busiest week of the year.	Laboratory manager	BLOCKING	Sizing of the server, the network and the interface.
Q-05	The acQuire product and version in use, and how assay results are loaded into it today.	Geology	BLOCKING	Defines the most important integration boundary in the system.
Q-06	Whether geology inserts its own control samples, at what rate, and using which certified reference materials.	Geology	HIGH	Determines the visibility rules, which touch every screen that displays a sample.
Q-07	What the server referred to as the end-of-day destination actually is, and what is stored on it.	IT	HIGH	May contain data that has to be migrated, or may be an existing system we have not accounted for.
Q-08	Whether any laboratory system exists today beyond spreadsheets, and what data must be brought across.	Laboratory / IT	HIGH	Migration scope.
Q-09	The current sample numbering convention, and whether it must be preserved.	Laboratory / Geology	HIGH	Identifier schemes are painful to change after go-live.
Q-10	Copies of a completed sample submission sheet, a batch sheet, a results report and a bar certificate.	Laboratory	HIGH	These four documents define the data model and the report templates more precisely than any discussion.
Q-11	The quality control rules as they are actually applied: tolerances, blank limits, duplicate limits and insertion rates.	Laboratory / QC	MEDIUM	Populates the initial configuration.
Q-12	Whether the fire assay finish is by AAS, gravimetric, or both, and whether parting is performed.	Laboratory	MEDIUM	Affects the assay workflow definition.
Q-13	Preparation detail: the splitting method used, and whether a screen or particle size check is performed on pulps.	Laboratory	MEDIUM	Preparation is where most real error originates and is usually the least instrumented step.
Q-14	Whether check or umpire samples are sent to an external laboratory, and how the comparison is handled today.	Laboratory / Geology	MEDIUM	Determines whether external laboratory handling is needed and when.

ID	WHAT WE NEED	OWNER	PRIORITY	WHY IT MATTERS
Q-15	Whether the operation pours refined bullion on site or doré for onward refining.	Metallurgy	MEDIUM	Decides whether refiner reconciliation is a real requirement.
Q-16	The policy on hosted or cloud systems, and any requirement that data remain in the country.	IT	MEDIUM	The design works either way, but the deployment plan and hardware differ.
Q-17	The record retention period the business requires.	Management / QC	MEDIUM	Storage sizing and archive policy.
Q-18	Site environment: server room, power stability, uninterruptible power supply, backup arrangements and who provides IT support on site.	IT	MEDIUM	Hardware specification and support model.
Q-19	The number of users by role, and the shift pattern worked.	Laboratory manager	LOW	Interface and licensing design.
Q-20	Whether further laboratory sites are planned.	Management	LOW	Determines when the central tier is needed.

THE ONE THAT MATTERS MOST

Until we know the make, model and output method of each instrument, integration effort is genuinely unknown — a single instrument can be two weeks of work or two months depending on how it was built and what its software will let us read. Every other unknown on this list affects sizing or sequencing. This one affects the price.

09

Confirmation

Signing below confirms that the process description in section 2 is accurate and that the requirements in section 5 represent what the laboratory needs, subject to the corrections marked on the document. It is not a commitment to cost or schedule; those follow once section 8 is answered.

LABORATORY MANAGER _____ name, signature, date	GEOLOGY _____ name, signature, date
INFORMATION TECHNOLOGY _____ name, signature, date	FOR THE SUPPLIER _____ name, signature, date

Draft issued for confirmation. Requirement, assumption and question identifiers are permanent and will be carried forward into the product requirements document unchanged. Withdrawn items are marked as withdrawn rather than renumbered.